

KRITIKA GUPTA

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SUMMARY

Computer Science graduate (MS, May 2026) with hands-on LLM application development experience, fine-tuning a DeepSeek LLM, implementing structured tool/function calling and agent orchestration, and incorporating RAG with vector database retrieval, built and evaluated in a real, shipped agentic system. Strong Python and SQL fluency, REST API integration experience, and cloud platform exposure (AWS). Comfortable working within engineering standards (Git, code review) and communicating clearly with both technical and non-technical stakeholders.

TECHNICAL SKILLS

LLM Application Development: HuggingFace Agents, DeepSeek LLM Fine-Tuning, Tool/Function Calling, Agent Orchestration, RAG, Vector Databases, Prompt Engineering

Languages: Python, Java, C, C++, SQL, JavaScript

Backend & APIs: Django, FastAPI, Flask, REST APIs, OOP

Data & SQL: SQL, Pandas, NumPy, Data Cleaning & Transformation, ETL Pipelines

Cloud: AWS (EC2, S3, Lambda, IAM), OCI, Docker

Engineering Fundamentals: Git, Code Review, Data Structures & Algorithms, Debugging, Documentation

PROFESSIONAL EXPERIENCE

Chicago Education Advocacy Cooperative (ChiEAC)

Jul 2026 - Present

Data Science Fellow / AI Data Analyst - Data Science Alliance

- Developing Generative AI and multi-agent capabilities, including LLM fine-tuning, RAG architectures, and multi-agent orchestration to improve operational efficiency.
- Designing scalable machine learning systems to transform data into predictive, interpretable models supporting organizational decision-making.

ThinkNext Technologies

Jun 2022 - Jul 2022

Software Engineering Intern

- Developed and maintained backend services for a production inventory management system using Python and Django.
- Designed, tested, and optimized SQL queries and database interactions across MySQL and Oracle for improved performance.
- Collaborated with the front-end team and participated in code review to ensure well-documented, maintainable code.

GTB Institute

Jul 2021 - Sep 2021

Python Programming Intern

- Used Pandas and NumPy for data analysis and visualization tasks.
- Led a team of 3 to deliver an airline management system end-to-end; received formal appreciation for leadership.

CMU Dining Services

Oct 2025 - May 2026

Team Member

- Delivered efficient customer service and collaborated with a diverse team in a fast-paced environment.

PROJECTS

Agentic AI Smart City Data Assistant

2026

Capstone

- Independently designed, built, and shipped an end-to-end agentic system, fine-tuning a DeepSeek LLM and implementing structured tool/function calling and agent orchestration for multi-step reasoning.
- Incorporated RAG and vector database retrieval to ground responses in relevant context rather than relying on prompting alone.
- Evaluated the system formally, tracking accuracy and reviewing failure cases, documenting findings and iterating on prompts and retrieval logic.

ETL Pipeline Development

Coursework

Academic Project

- Built ETL pipelines to extract, clean, transform, and load structured data for downstream analysis, working with SQL-based storage.

Cloud-Based URL Shortener with AWS

2024

Independent

- Designed, built, and deployed a full-stack, serverless service with a REST API, integrating multiple AWS services (EC2, S3, Lambda, DynamoDB).
- Regime-Aware Deep Learning Framework for Predictive Maintenance**
ML Research

2025
- Co-authored a hybrid deep learning model for aircraft engine failure prediction, applying rigorous statistical validation and benchmarking against multiple baseline models.
- Interactive Sales Dashboard**
Academic

2023
- Built a Tableau dashboard with drill-down capabilities and dynamic filters to visualize key business KPIs for non-technical stakeholders, directly relevant experience with retail-style KPI reporting.

PUBLICATIONS

Gupta, K., Barot, N., & Gargees, R. S. (2026). *Regime-Aware Deep Learning Framework for Predictive Maintenance in Industrial Systems*. MLISE 2026. DOI: 10.1109/MLISE70044.2026.11607667

EDUCATION

Central Michigan University <i>MS, Computer Science GPA: 3.77/4.0</i>	May 2026
DAV University, India <i>BTech, Computer Science GPA: 3.07/4.0</i>	Jun 2023

CERTIFICATIONS

- Hugging Face Agentic AI: Agent orchestration, tool calling, LLM reasoning - 2025
- NVIDIA Deep Learning Workshop: Neural networks, optimization, NVIDIA frameworks - 2025

HONORS

- Team Leader, Capstone AI Project
- Departmental Scholarship - CMU
- Leadership Award - GTB Institute
- Bronze Medal, National Science Olympiad